

Diego M. Salinas

AI/ML/CV & Full Stack Engineer

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Professional Summary

Experienced **AI/ML engineer and full-stack software developer** specializing in **AI model development and integration, computer vision, real-time control, robotics, and embedded AI systems**. Skilled at building **low-latency, multi-threaded pipelines, AI inference for sensor/camera streams**, and **full-stack integration**. Proven track record in delivering **production-ready, safety-focused, and high-performance systems** across cloud and edge environments.

Professional Experience

Applied Intuition – AI/CV Engineer (Remote)

Oct 2023 – June 2025

- Led development and deployment of **real-time computer vision and multimodal AI pipelines** for simulation and embedded systems.
- Built **multi-stream RTSP pipelines** for **vehicle detection and license plate recognition** on NVIDIA Jetson devices, delivering **real-time structured webhook data**.
- Designed **low-latency inference systems** for **multi-camera restaurant monitoring**, integrating people counting, cleanliness detection, worker behavior analysis, and audio anomaly detection.
- Developed **FastAPI-based model serving infrastructure** with **Prometheus & Grafana** monitoring.
- Optimized deep learning workloads using **ONNX Runtime, TensorRT, and Triton Inference Server** for edge deployment.

Xometry (via DataTalent Solutions) – AI/ML/CV Engineer(Remote)

Mar 2022 – Sep 2023

- Built **time-series and predictive AI models** for manufacturing demand forecasting.
 - Developed **data preprocessing and inference pipelines** for large-scale sensor and operational datasets.
 - Delivered **interactive dashboards** for stakeholders to visualize predictions in real-time.
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Key Projects & Achievements

Ultimate ALPR System (Jetson Devices)

- Designed a **multi-stream vehicle and license plate recognition system** processing **10+ RTSP streams simultaneously**.
- Delivered **real-time structured webhook data** to enterprise systems for access control and traffic analytics.
- Achieved **high accuracy (>95% detection and OCR)** on license plates under varying lighting and motion conditions.
Tech: NVIDIA Jetson, PyTorch, OpenCV, YOLOv8, ONNX, TensorRT, FastAPI

Restaurant Monitoring System

- Developed a **multi-camera real-time monitoring platform** for restaurants.
- Detected people, cleanliness issues, and worker behavior; integrated **audio-based anomaly alerts** for enhanced monitoring.
- Scalable design allows **processing multiple RTSP streams concurrently** with low-latency inference.
Tech: PyTorch, OpenCV, YOLOv8, Multi-threading, Docker, FastAPI

Watermark Removal Pipeline

- Automated **video watermark removal** for hundreds of videos using **LLaMA-based inpainting models**.
- Designed a **batch processing pipeline** for distributed GPU execution, reducing processing time by 70%.

Tech: PyTorch, LLaMA Inpainting, OpenCV, Docker

OCR-Based Handwritten Document Extraction

- Built **high-accuracy OCR system** for challenging handwritten text captured by camera.
- Combined **vision-language models** with custom preprocessing for improved text recognition.
- Delivered structured JSON outputs for downstream workflows.

Tech: PyTorch, Qwen-VL, OpenCV, Python

Home Security & Thief Detection System

- Multi-camera surveillance with **face recognition** and **sound-based intrusion alerts**.
- Optimized for **offline inference** on embedded devices; low-latency alerting system (<1s reaction).

Tech: PyTorch, OpenCV, TensorRT, Raspberry Pi, Jetson

Home Assistant Energy Optimization

- AI-driven system predicting room occupancy and controlling temperature automatically.
- Reduced energy consumption by **20–25%** in tested deployments.
- Integrated with **IoT sensors and microcontrollers** for real-time control.

Tech: Python, TensorFlow, Raspberry Pi, IoT Integration, Signal Processing

Technical Skills

Programming & Development:

C/C++, Python, C#, Java, JavaScript, Node.js, React.js, Full-Stack, Desktop Apps

AI/ML & Computer Vision:

PyTorch, TensorFlow, OpenCV, YOLOv8, LLaVA, CLIP, Multimodal AI, LLMs, RAG

Embedded & Robotics:

Raspberry Pi, NVIDIA Jetson, Sensor Integration, Low-Latency Pipelines, Real-Time Robotics

Deployment & Optimization:

Docker, TensorFlow Lite, PyTorch Mobile, ONNX, CI/CD Pipelines, Model Quantization

Automation & Tools:

Signal Processing, Multi-Threading, API Integration, Real-Time Monitoring

Standards & Compliance:

ISO 13485, IEC 62304

Highlights & Certifications

- Extensive experience in **real-time video and sensor fusion**
 - Expertise in **low-latency control software** and **AI inference pipelines**
 - Proven success in **embedded AI deployment** on constrained devices
 - Familiarity with **medical device safety and compliance** frameworks
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Education**B.S. in Computer Engineering**

University of Texas at Dallas – May 2019